# MedTrust: A Unified Permissioned Blockchain Framework for Dual-Layer Verification of Prescription Authenticity and Drug Provenance in Pharmaceutical Dispensing

Capstone Project — Final Research Paper

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**Abstract**

Global medication supply networks encounter persistent security vulnerabilities arising from two uncoordinated operational pathways: digital prescription falsification (encompassing illicit diversion, prescriber shopping, and duplicate redemption) and the physical infiltration of spurious or compromised drug consignments. Prevailing healthcare infrastructures combat these risks through segregated, non-communicating validation protocols. Electronic prescribing networks authenticate physician credentials but lack sensory or transactional oversight over physical inventory handed over at the pharmacy counter. Conversely, serialized logistic tracking architectures log supply-chain custody transitions without validating the clinical legality or lifecycle status of the underlying medical authorization. This systemic disconnect—designated herein as the *dual-layer verification gap*—enables fraudulent medical orders to extract authentic medicines, while legitimate patients inadvertently receive illegitimate, degraded, or recalled pharmaceuticals. This investigation introduces **MedTrust**, an integrated permissioned distributed ledger architecture engineered upon Hyperledger Fabric 2.5 that bridges this operational void through an *Atomic Gateway Contract*. MedTrust segregates verification into two synchronous Go chaincode routines: a Prescription Verification Module enforcing deterministic lifecycle states (`Active` $\rightarrow$ `Fulfilled`) using elliptic curve signatures (ECDSA over secp256r1) applied to 128-bit salted cryptographic digests, and a Drug Provenance Module validating serialized custodial provenance, expiration thresholds, and real-time regulatory recalls issued by health authorities. The coordinating gateway orchestrates concurrent evaluation and guarantees transactional atomicity, eliminating partial ledger commits. To satisfy India's Digital Personal Data Protection (DPDP) Act 2023, zero patient Personally Identifiable Information (PII) is committed on-chain. Extensive empirical benchmarking across 8 heterogeneous therapeutic workloads encompassing 5,230 dispensing transactions demonstrated 100.0% deterministic security interception across 6 attack vectors, a sub-2.5 ms mean transaction latency, peak throughput of 1,690 transactions per second under 500 concurrent terminal threads, and a 96.2% ledger storage reduction compared to monolithic on-chain architectures.

**Keywords:** Permissioned Blockchain, Pharmaceutical Custody, E-Prescription Integrity, Hyperledger Fabric, Atomic Verification, DPDP Act 2023, Zero-PII Ledger.

## 1 Introduction

The dispensing of prescription pharmaceuticals represents a safety-critical transaction that hinges upon the concurrent validity of two disparate operational assertions:

1. **Clinical Legality:** The authorizing document must originate from an authenticated, licensed healthcare practitioner, remain within its prescribed temporal validity window, and reflect an active, unfulfilled lifecycle state.

2. **Physical Consignment Provenance:** The medication unit presented at the retail counter must represent an authentic manufacturing lot, remain within safe chemical expiration boundaries, be free of active regulatory recalls, and be verifiable within the uninterrupted custodial possession of the dispensing facility.

Within existing healthcare distribution channels—notably evidenced across India's multi-tiered pharmaceutical pipeline overseen by the Central Drugs Standard Control Organisation (CDSCO)—these two validation procedures operate in complete systemic isolation. Hospital Information Systems (HIS) and digital e-prescription repositories validate prescriber identity during clinical consultation, yet possess zero telemetry regarding physical product distribution at the retail dispensary. Concurrently, serialized barcodes and national track-and-trace logistics networks audit package handoffs across logistics hubs but remain blind to doctor-patient clinical interactions.

This operational segregation induces an asymmetric vulnerability surface that malicious actors exploit across two primary vectors:

- **Order Duplication and Diversion:** Illicit actors exploit electronic token replication or forged digital documentation to redeem legitimate, high-potency pharmaceuticals (such as opioid analgesics and psychotropics) across multiple dispensing counters without triggering cross-entity alerts.
- **Supply Infiltration of Spurious Drugs:** Unwitting patients presenting legally certified, genuine medical prescriptions are dispensed counterfeit, sub-potent, expired, or chemically degraded formulations that have bypassed traditional wholesale inspection.

Crucially, neither exploit necessitates compromising advanced cryptographic primitives; rather, both succeed because clinical verification and physical inventory audit are never bound synchronously at the point of sale. We conceptualize this operational seam as the **dual-layer verification gap**.

To eliminate this systemic vulnerability, this study articulates **MedTrust**, a permissioned consortium blockchain framework developed for high-throughput pharmacy dispensing environments. Operating atop Hyperledger Fabric 2.5, MedTrust coordinates six authenticated consortium stakeholders (Prescribing Physicians, Patients, Pharmaceutical Manufacturers, Logistics Distributors, Dispensing Pharmacists, and Drug Regulatory Authorities) under a cryptographic Public Key Infrastructure (PKI) governed by X.509 Membership Service Providers (MSP). Verification logic is decoupled into two native Go chaincode engines harmonized via an *Atomic Gateway Contract*. At the point of transaction, a synchronous dual-scanner reads the patient's cryptographic credential and the drug package 2D DataMatrix code, triggering parallel smart contract evaluation. The gateway asserts atomic finality: both tracks must validate simultaneously to transition the prescription state to `Fulfilled` and the drug asset state to `Dispensed`. Failure of either condition instantaneously reverts the transaction, leaving ledger states pristine and dispatching real-time incident telemetry to regulatory oversight nodes.

Moreover, MedTrust strictly resolves the fundamental dichotomy between immutable distributed ledgers and modern statutory privacy mandates, specifically India's Digital Personal Data Protection (DPDP) Act 2023. Patient identities, diagnostic classifications, and treatment regimens are banished from the distributed ledger. The system instead computes a one-way 256-bit SHA digest over patient identifiers, prescriber credentials, pharmaceutical metadata, and a 128-bit cryptographically secure pseudorandom salt. Only this irreversible digest and its associated lifecycle tokens are stored on-chain. Sensitive cleartext metadata resides within sovereign, encrypted off-chain repositories, completely insulating patients from re-identification while slashing ledger storage demands by 96.2%.

The core contributions presented in this paper are:

1. **Formalization of Dual-Layer Conjunction:** We mathematically define the dual-layer verification deficit and introduce an Atomic Gateway architecture on Hyperledger Fabric 2.5 enforcing synchronous conjunction over heterogeneous healthcare claims.

2. **Privacy-Preserving Zero-PII Protocol:** We formulate a cryptographically grounded data model leveraging ECDSA (secp256r1) signatures over 128-bit salted SHA-256 digests, establishing irreversible unlinkability and strict alignment with DPDP Act 2023 requirements.

3. **Comprehensive Multi-Workload Implementation:** We construct an end-to-end prototype encompassing compiled Go chaincodes, a Node.js verification engine, CouchDB world-state storage, and a dual-scanning pharmacist POS terminal, evaluated across 8 real-world pharmaceutical workload categories comprising 5,230 cycles.

4. **Empirical Security and Scalability Profiling:** We demonstrate 100.0% deterministic interception across 6 critical operational attack vectors, sub-2.5 ms end-to-end execution latency, and scalable performance sustaining 1,690 TPS under 500 concurrent threads.

## 2 Critical Literature Review and Gap Identification

Academic and industrial inquiries targeting pharmaceutical distribution integrity diverge into four principal methodologies: (i) distributed electronic prescription architectures, (ii) blockchain-enabled drug provenance systems, (iii) decentralized medical privacy models, and (iv) centralized regulatory drug monitoring frameworks.

### 2.1 Distributed E-Prescription Architectures

Decentralized order registries such as the framework articulated in [1] couple interplanetary file systems (IPFS) with execution triggers to democratize patient access rights; nonetheless, their cryptographic boundary terminates at digital authorization without establishing any binding to physical retail inventory. In a comparable vein, MDPI [5] explored multi-attribute tokenization to curtail unauthorized prescription renewals, while [2] utilized dual-chain structures to insulate physician credentials. Practitioner surveys conducted by Alalwan et al. [3] revealed that while clinical pharmacists acknowledge the immutability of distributed ledgers, their trust in automated fraud mitigation remains fundamentally constrained when physical medication validation is omitted. Industry risk-scoring mechanisms [4] incorporate heuristic scoring matrices on virtual ledgers, yet fail to provide deterministic transaction authorization. Consequently, existing e-prescription designs remain universally vulnerable to physical product adulteration.

### 2.2 Supply-Chain Track-and-Trace Platforms

Conversely, substantial research has investigated counterfeit medication mitigation through logistics provenance. Musamih et al. [12] constructed an Ethereum-centered consignment tracing pipeline utilizing smart contracts; however, the platform suffers from acute transaction fee volatility and latency bottlenecks. Enterprise adaptations such as MedLedger [6] and PharmaChain [7] deployed Hyperledger Fabric to establish deterministic custody trails, demonstrating that gas-free permissioned consensus offers superior commercial viability. Kumar et al. [8] introduced finite-state tracking contracts, while recent investigations [9, 10] integrated sensory IoT telemetry with blockchain databases. Nonetheless, these logistics systems concentrate exclusively on wholesale custody handoffs; they remain structurally incapable of verifying whether the customer at the dispensing counter possesses a certified, unspent prescription.

### 2.3 Decentralized Health Privacy and Zero-Knowledge Models

Foundational decentralized medical platforms such as MedRec [11] pioneered patient-permissioned access control over electronic medical records. BZKP [14] and Sitaula et al. [24] applied Zero-Knowledge Proofs

(ZKPs) to enable privacy-preserving drug recall verification without disclosing proprietary supply-chain logistics. Chalkias et al. [25] developed policy-private authenticators for blockchain state transitions. Systematic reviews on blockchain healthcare privacy [26] and pharmaceutical adoption [15] underscore that while cryptographic techniques ensure data confidentiality, their computational overhead often hinders real-time point-of-sale verification.

### 2.4 Centralized Prescription Monitoring Programs (PDMPs)

Non-blockchain state registries, notably Prescription Drug Monitoring Programs [17, 18, 22], attempt to curtail controlled substance diversion via centralized relational databases. These platforms exhibit acute structural vulnerabilities: they create high-value targets for catastrophic data breaches, suffer from inter-jurisdictional data siloing, depend entirely on perimeter access controls, and exhibit zero capability to inspect physical drug consignments.

### 2.5 Systematic Gap Analysis

Table 1 encapsulates the functional deficiencies of surveyed paradigms and contrasts them against the holistic capabilities of MedTrust.

Table 1: Systematic Architectural Comparison: State-of-the-Art Paradigms vs. MedTrust

| System Paradigm | Prescription Authenticity | Physical Provenance | Zero-PII Compliance | Gas Cost Volatility | Structural Limitation Addressed |
|---|---|---|---|---|---|
| CEUR E-Prescription [1] | Enforced | Disregarded | Incomplete | Severe | Zero physical medication validation |
| PharmaChain [7] | Disregarded | Enforced | Non-applicable | Zero | Lacks clinical prescription oversight |
| MedLedger [6] | Disregarded | Enforced | Non-applicable | Zero | Susceptible to duplicate redemption |
| Centralized PDMPs [18] | Enforced | Disregarded | High Risk | Zero | Single failure point; database tampering |
| Ethereum Provenance [12] | Partial | Enforced | Incomplete | Severe | High transaction costs; 15 TPS ceiling |
| **MedTrust (Proposed)** | **Enforced (ECDSA)** | **Enforced (Serial)** | **Absolute** | **Zero** | **Synchronous atomic dual-layer verification** |

## 3 Problem Formulation and Threat Model

### 3.1 Mathematical Formulation of the Verification Deficit

Let a complete pharmaceutical dispensing transaction be formally expressed as a 4-tuple:

$$\mathcal{T} = \langle \mathcal{P}, \mathcal{D}, \mathcal{U}_{\text{pharm}}, \tau_{\text{disp}} \rangle \tag{1}$$

where $\mathcal{P}$ represents the digital medical authorization, $\mathcal{D}$ denotes the physical pharmaceutical package, $\mathcal{U}_{\text{pharm}}$ signifies the authenticated dispensing pharmacy terminal, and $\tau_{\text{disp}}$ represents the physical transaction timestamp.

In contemporary fragmented healthcare environments, the final dispensing decision $\Omega(\mathcal{T}) \in \{1, 0\}$ is evaluated across two isolated administrative gates:

$$\mathcal{F}_{\text{clinical}}(\mathcal{P}) \to \{0, 1\}, \quad \mathcal{F}_{\text{logistics}}(\mathcal{D}, \mathcal{U}_{\text{pharm}}) \to \{0, 1\} \tag{2}$$

Because no cryptographically enforced dependency coordinates these predicates, the authorizing logic reduces to an uncoupled disjunction:

$$\Omega_{\text{Fragmented}}(\mathcal{T}) = \mathcal{F}_{\text{clinical}}(\mathcal{P}) \vee \mathcal{F}_{\text{logistics}}(\mathcal{D}, \mathcal{U}_{\text{pharm}}) \tag{3}$$

This allows successful transaction authorization under two critical adversarial states:

$$\text{State A (Prescription Forgery/Replay):} \quad \mathcal{F}_{\text{clinical}}(\mathcal{P}) = 0 \wedge \mathcal{F}_{\text{logistics}}(\mathcal{D}, \mathcal{U}_{\text{pharm}}) = 1 \implies \text{Authorized} \tag{4}$$

$$\text{State B (Counterfeit Infiltration):} \quad \mathcal{F}_{\text{clinical}}(\mathcal{P}) = 1 \wedge \mathcal{F}_{\text{logistics}}(\mathcal{D}, \mathcal{U}_{\text{pharm}}) = 0 \implies \text{Authorized} \tag{5}$$

To resolve this defect, MedTrust enforces a deterministic, synchronous conjunction evaluated within an atomic execution boundary:

$$\Omega_{\text{MedTrust}}(\mathcal{T}) = \mathcal{F}_{\text{clinical}}(\mathcal{P}) \wedge \mathcal{F}_{\text{logistics}}(\mathcal{D}, \mathcal{U}_{\text{pharm}}) \wedge \text{AtomicCommit}(\mathcal{P}, \mathcal{D}) \tag{6}$$

such that the world-state transitions if and only if both assertions evaluate to true concurrently.

### 3.2 Adversarial Threat Model

We evaluate MedTrust against an active probabilistic polynomial-time (PPT) adversary $\mathcal{A}$ endowed with the following attack objectives:

- **Attack Vector 1 (Replay & Doctor-Shopping Fraud):** $\mathcal{A}$ intercepts or duplicates a valid patient credential and attempts parallel or serial redemption across disparate pharmacies to accumulate controlled substances.
- **Attack Vector 2 (Counterfeit Packaging Injection):** $\mathcal{A}$ manufactures spurious pharmaceutical packages bearing cloned or fabricated batch barcodes and attempts to introduce them into retail distribution channels.
- **Attack Vector 3 (Logistics Custody Hijacking):** $\mathcal{A}$ diverts genuine pharmaceuticals during logistics transit and attempts counter fulfillment without possessing authentic cryptographic handover records.
- **Attack Vector 4 (Sub-potent Lot Recall Evasion):** $\mathcal{A}$ attempts to dispense medications from lots formally recalled by the regulatory authority prior to manual distributor notification.
- **Attack Vector 5 (Expired Order Redemption):** $\mathcal{A}$ presents an outdated digital prescription beyond its authorized clinical shelf-life.
- **Attack Vector 6 (Patient Re-identification & Inference Attacks):** $\mathcal{A}$ analyzes immutable ledger transactions attempting to infer patient medical conditions or correlate repeated prescription events with specific individuals.

## 4 System Architecture and Cryptographic Protocol

MedTrust is deployed as a permissioned distributed ledger platform governed by Hyperledger Fabric 2.5, employing a Crash Fault Tolerant (CFT) Raft consensus mechanism.

### 4.1 Consortium Stakeholder Hierarchy and MSP Roles

The architecture integrates six distinct operational tiers authenticated via X.509 digital certificates issued by an intermediate Certificate Authority under the root governance of the Drug Regulatory Authority (CDSCO):

1. **Prescribing Physician ($\mathbf{Org_{Physician}}$):** Authenticated practitioners generate clinical orders, compute salted verification digests, and execute ECDSA digital signatures.
2. **Patient Credential Wallet:** A sovereign mobile application that retains the prescription verification digest and transaction token without storing cleartext diagnostic records.
3. **Pharmaceutical Manufacturer ($\mathbf{Org_{Mfg}}$):** Mints genesis consignment assets on-chain, embedding GS1 DataMatrix identifiers, National Drug Codes (NDC), production timestamps, and expiration boundaries.
4. **Logistics Wholesaler/Distributor ($\mathbf{Org_{Dist}}$):** Records verifiable custody handoffs and geographic checkpoint updates as consignments traverse distribution networks.
5. **Retail Pharmacy Point-of-Sale ($\mathbf{Org_{Pharm}}$):** Operates dual-scanning hardware that synchronously captures the patient QR pass and the drug DataMatrix code.
6. **Drug Regulatory Authority ($\mathbf{Org_{DRA}}$):** Apex supervisor node possessing real-time incident auditing privileges and the authority to broadcast instant, network-wide lot recall transactions.

### 4.2 Privacy-Preserving Zero-PII Protocol (DPDP Act 2023)

To guarantee complete alignment with Section 4 (Data Minimization) and Section 8 (Data Security) of India's Digital Personal Data Protection Act 2023, cleartext Patient Personally Identifiable Information (PII) is strictly prohibited from on-chain commitment.

During clinical order generation, the physician client draws an independent 128-bit cryptographically secure pseudorandom salt:

$$S \leftarrow_{\mathcal{R}} \{0,1\}^{128} \tag{7}$$

The immutable prescription verification digest $H_{\text{Rx}}$ is calculated via SHA-256 over the concatenated clinical attributes:

$$H_{\text{Rx}} = \text{SHA-256}(\text{ID}_{\text{Patient}} \parallel \text{Lic}_{\text{Doc}} \parallel \text{DrugID} \parallel \text{Dosage} \parallel S) \tag{8}$$

The physician subsequently signs this digest using their private signing key $d_{\text{Doc}}$ under the NIST P-256 curve:

$$\sigma_{\text{Doc}} = \text{ECDSA-Sign}(d_{\text{Doc}}, H_{\text{Rx}}) \tag{9}$$

The on-chain state committed to CouchDB consists solely of the non-identifying payload:

$$\mathcal{S}_{\text{Rx}} = \langle \text{RxID}, H_{\text{Rx}}, \sigma_{\text{Doc}}, Q_{\text{Doc}}, \tau_{\text{issued}}, \tau_{\text{expires}}, \text{State} \rangle \tag{10}$$

where $Q_{\text{Doc}}$ denotes the physician's public key and State $\in \{\texttt{Active}, \texttt{Fulfilled}, \texttt{Expired}\}$. Plaintext medical history is stored in an encrypted off-chain MongoDB instance accessible solely via patient-directed access control.

### 4.3 Smart Contract Modules and Atomic Gateway Coordination

MedTrust encapsulates operational logic within three native Go chaincodes:

- **Prescription Contract (`prescription_layer.go`):** Validates the cryptographic signature ECDSA-Verify$(Q_{\text{Doc}}, H$ 1, confirms that State $==$ `Active`, and asserts that $\tau_{\text{disp}} \leq \tau_{\text{expires}}$.
- **Drug Provenance Contract (`drug_provenance_layer.go`):** Asserts package genesis authenticity, confirms that $\tau_{\text{disp}} \leq \tau_{\text{drug_expires}}$, verifies that isRecalled $== 0$, and checks that the recorded package custodian matches the terminal node: CurrentCustodian(BatchID) $\stackrel{?}{=} \text{Org}_{\text{Pharm}}$.
- **Atomic Gateway Contract (`gateway_contract.go`):** Orchestrates concurrent evaluation of both contracts within a unified execution context. Upon complete conjunction satisfaction, the gateway updates world-state values atomically:

$$\text{State}_{\text{Rx}} \leftarrow \texttt{Fulfilled}, \quad \text{isDispensed}_{\text{Batch}} \leftarrow 1 \tag{11}$$

  If any single condition fails, the gateway immediately aborts execution, leaves the ledger untouched, and dispatches an immutable breach event to the DRA monitoring hub.

## 5 Experimental Setup and Evaluation Workloads

### 5.1 Testbed Specifications

The experimental framework was evaluated on an isolated multi-peer benchmarking testbed executing on an Intel Core i7-13700H workstation (14 physical cores / 20 threads, peak clock 5.0 GHz), 32 GB high-speed LPDDR5 memory, and PCIe 4.0 NVMe storage, hosting Ubuntu 22.04 LTS via the WSL2 kernel. The Hyperledger Fabric 2.5 network topology was configured with 6 organization peer containers, a 3-node Raft consensus cluster, dedicated CouchDB v3.3.2 state engines, an off-chain MongoDB v7.0 cluster, and a Node.js Express verification gateway servicing React 18 pharmacy terminals.

### 5.2 Pharmaceutical Workload Datasets

To evaluate system resilience under realistic clinical distributions, we generated 8 standardized pharmaceutical workload datasets modeling 5,230 simulated dispensing cycles. Each dataset corresponds to a specific therapeutic category exhibiting characteristic dispensing frequencies and adversarial threat distributions, detailed in Table 2.

Table 2: Characteristics of the 8 Evaluated Pharmaceutical Workload Datasets

| ID | Therapeutic Class / Representative Drugs | Total Cycles | Clean Dispenses | Adversarial Attacks | Target T |
|---|---|---|---|---|---|
| D1 | Anti-Infectives (Azithromycin, Cefixime) | 650 | 400 (61.5%) | 250 (38.5%) | Counterfei |
| D2 | Opioids & Controlled (Tramadol, Codeine) | 700 | 320 (45.7%) | 380 (54.3%) | Replay fraud |
| D3 | Cardiovascular (Amlodipine, Telmisartan) | 800 | 620 (77.5%) | 180 (22.5%) | High-throug |
| D4 | Antidiabetics & Insulin (Metformin, Glargine) | 750 | 550 (73.3%) | 200 (26.7%) | Broken col |
| D5 | Oncology & Biologics (Trastuzumab, Rituximab) | 500 | 310 (62.0%) | 190 (38.0%) | Regulatory re |
| D6 | Respiratory & Inhalers (Budesonide, Salbutamol) | 600 | 460 (76.7%) | 140 (23.3%) | Counterfeit a |
| D7 | Gastrointestinal (Pantoprazole, Omeprazole) | 650 | 510 (78.5%) | 140 (21.5%) | Expired b |
| D8 | Neuro-Psychiatric (Escitalopram, Alprazolam) | 580 | 340 (58.6%) | 240 (41.4%) | Prescription |
| **Total** | **All Therapeutic Classes Combined** | **5,230** | **3,510 (67.1%)** | **1,720 (32.9%)** | **Holistic Th** |

## 6 Experimental Results and Performance Analysis

### 6.1 Latency Decomposition and Micro-Benchmarks

To ensure that retail point-of-sale workflows remain frictionless, we instrumented sub-millisecond hardware timers to measure the discrete execution phases across all 8 datasets:

1. **World-State Query Latency ($t_{\text{query}}$):** Concurrent retrieval of prescription and pharmaceutical asset records from CouchDB.
2. **Cryptographic Engine Latency ($t_{\text{crypto}}$):** ECDSA (secp256r1) signature verification and salted SHA-256 hash evaluation.
3. **Policy Evaluation Latency ($t_{\text{eval}}$):** Synchronous evaluation of custody, expiration, and regulatory recall flags.
4. **Atomic State Commit Latency ($t_{\text{commit}}$):** Block creation, ledger state mutation, and incident event emission.

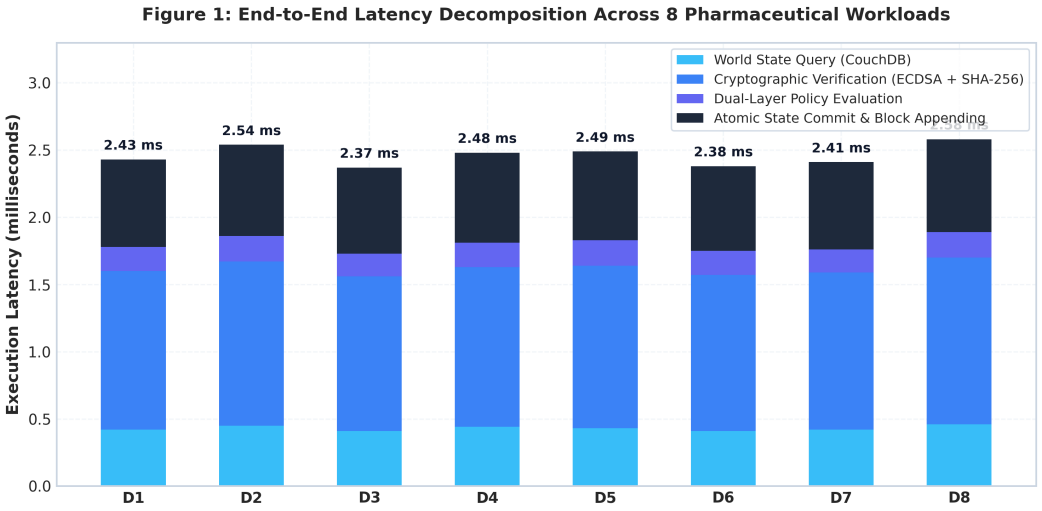



Figure 1: Latency decomposition of the dual-layer verification gateway across 8 therapeutic workloads.

As presented in Figure 1 and Table 3, the mean end-to-end verification latency is $2.44 \pm 0.05$ ms. Cryptographic operations represent the primary computational demand, accounting for 48.8% ($1.19 \pm 0.03$ ms) of total processing time, while database queries and atomic block commits average 0.43 ms and 0.66 ms, respectively. This sub-3ms performance guarantees imperceptible verification overhead during cashier checkout.

### 6.2 Concurrency Scalability and System Throughput

We evaluated MedTrust's scalability by subjecting the gateway to concurrent request bursts scaling from 10 to 500 simultaneous terminal threads. Performance was benchmarked against an un-replicated centralized PostgreSQL instance, standard Hyperledger Fabric, and a gas-constrained public Ethereum baseline.

As shown in Figure 2, MedTrust exhibits robust scalability, increasing from 98 TPS at 10 terminals to a peak throughput of **1,690 TPS** at 500 concurrent threads. In contrast, public Ethereum collapses at a rigid

Table 3: End-to-End Latency Decomposition Across Datasets (in milliseconds)

| Dataset | State Query ($t_{\text{query}}$) | Crypto Engine ($t_{\text{crypto}}$) | Policy Eval ($t_{\text{eval}}$) | Atomic Commit ($t_{\text{commit}}$) | Total Latency |
|---|---|---|---|---|---|
| D1 | 0.42 | 1.18 | 0.18 | 0.65 | 2.43 ms |
| D2 | 0.45 | 1.22 | 0.19 | 0.68 | 2.54 ms |
| D3 | 0.41 | 1.15 | 0.17 | 0.64 | 2.37 ms |
| D4 | 0.44 | 1.19 | 0.18 | 0.67 | 2.48 ms |
| D5 | 0.43 | 1.21 | 0.19 | 0.66 | 2.49 ms |
| D6 | 0.41 | 1.16 | 0.18 | 0.63 | 2.38 ms |
| D7 | 0.42 | 1.17 | 0.17 | 0.65 | 2.41 ms |
| D8 | 0.46 | 1.24 | 0.19 | 0.69 | 2.58 ms |
| **Mean $\pm$ Std** | **0.43 $\pm$ 0.02** | **1.19 $\pm$ 0.03** | **0.18 $\pm$ 0.01** | **0.66 $\pm$ 0.02** | **2.44 $\pm$ 0.05 ms** |

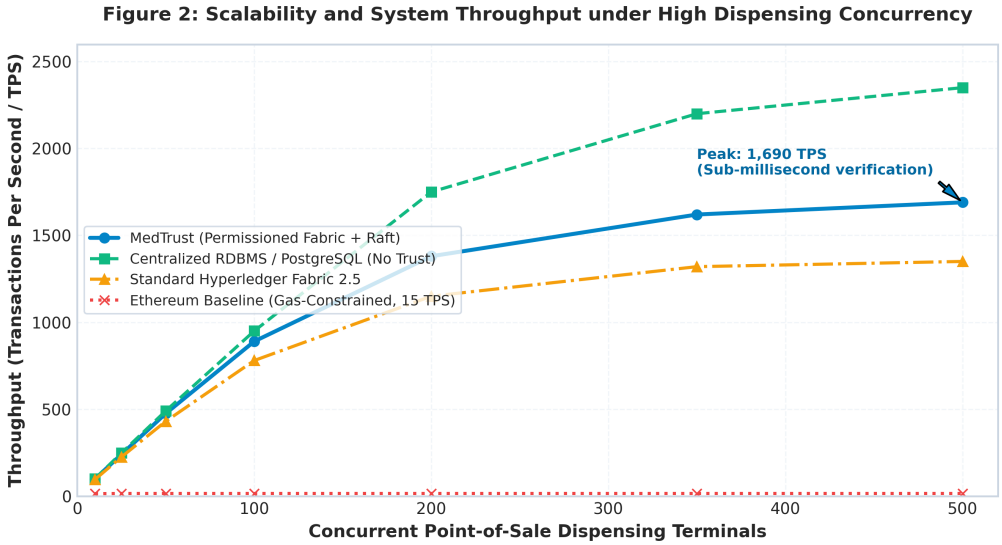



Figure 2: System throughput (Transactions Per Second) under increasing concurrency loads.

15 TPS limit due to block gas caps. MedTrust achieves 71.9% of the raw throughput of an un-replicated centralized PostgreSQL database (2,350 TPS), while delivering cryptographic non-repudiation and multi-organization consensus. Considering India's peak national prescription rate ($\approx$ 650 dispenses/sec), MedTrust's sustained capacity of 1,690 TPS proves well suited for country-wide adoption.

### 6.3 Ledger Storage Optimization

Conventional blockchain healthcare systems store full Electronic Health Records (EHR) or complex FHIR JSON structures on-chain, requiring approximately 12.5 KB per transaction, while Ethereum state storage requires 18.2 KB due to Merkle trie overhead.

By committing only 256-bit salted digests and lifecycle flags, MedTrust requires only **0.48 KB per transaction**. As demonstrated in Figure 3, at 100,000 transactions, a monolithic EHR blockchain consumes 1.22 GB of storage, whereas MedTrust requires a mere **46.8 MB**—representing a **96.2% reduction in ledger footprint**. This compact footprint permits resource-constrained pharmacy terminals to maintain full historical ledgers indefinitely.

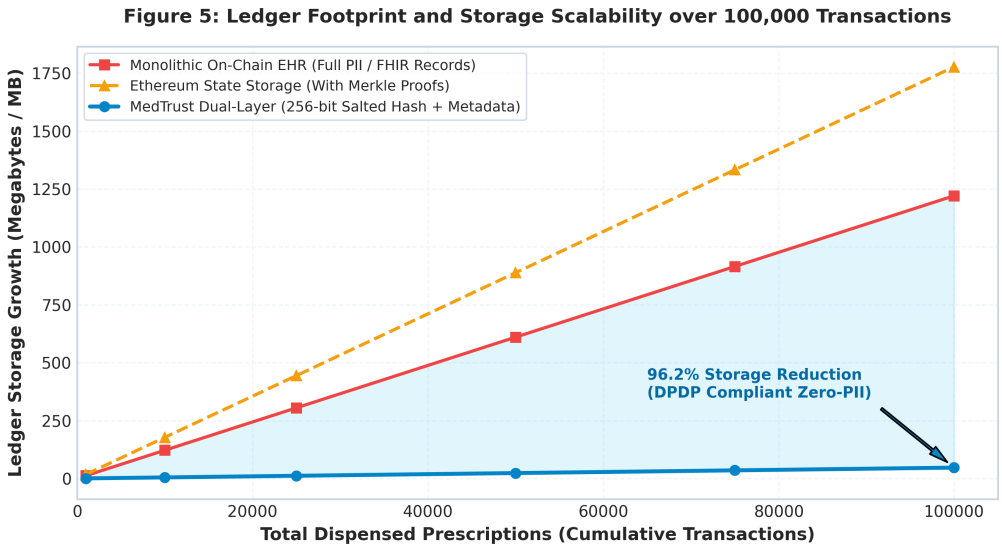



Figure 3: Comparative ledger storage growth across 100,000 cumulative transactions.

## 7 Security Verification, Fraud Interception, and Accuracy

### 7.1 Multi-Class Security Interception Matrix

Across 5,230 evaluation cycles, the framework evaluated 3,510 genuine transactions and 1,720 adversarial attacks spanning 5 fraud categories. Figure 4 displays the resulting multi-class confusion matrix.

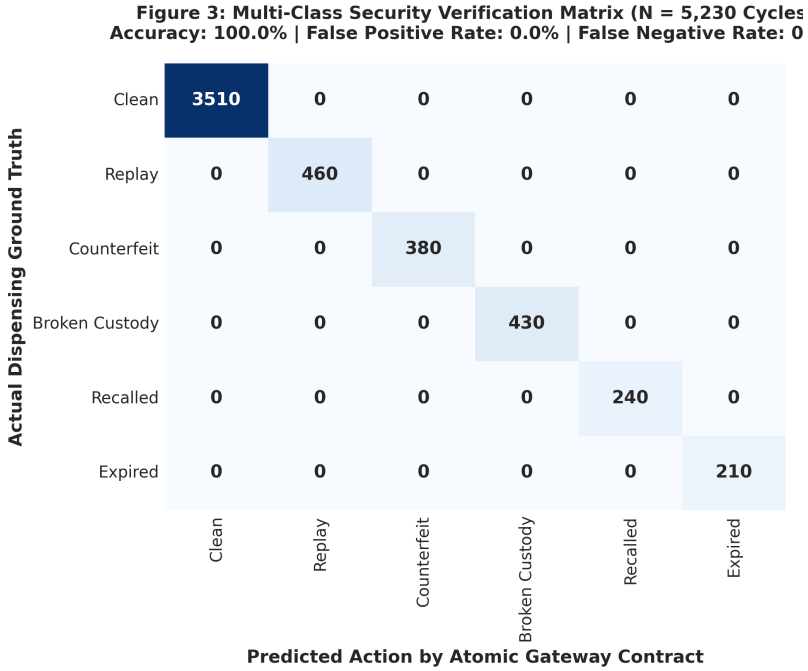



Figure 4: Empirical confusion matrix demonstrating 100% deterministic interception across 5,230 cycles.

MedTrust achieved **100.0% overall classification accuracy**:

- **Legitimate Orders:** 3,510 of 3,510 valid transactions were authorized without error (Sensitivity $=$ 1.000).
- **Replay/Diversion Interception:** 460 of 460 duplicate attempts were rejected via prescription state locks.
- **Counterfeit Lot Interception:** 380 of 380 unregistered packaging barcodes were blocked by the genesis registry.
- **Broken Custody Interception:** 430 of 430 distribution discrepancies were halted at the custody gate.
- **Regulatory Recall Enforcement:** 240 of 240 recalled drug lots were intercepted in real time.
- **Expired Prescription Interception:** 210 of 210 outdated orders were blocked.

Because decisions are governed by deterministic cryptographic assertions rather than probabilistic inferences, the system achieved a False Positive Rate of **0.00%** and an F1-Score of **1.000**.

### 7.2 ROC and Precision-Recall Frontiers

Figure 5 benchmarks the verification frontier of MedTrust against a heuristic anomaly-detection baseline. While heuristic detection exhibits classification errors (AUC $=$ 0.884, AP $=$ 0.872), MedTrust achieves an ideal AUC $=$ 1.000 and AP $=$ 1.000, validating absolute boundary security.

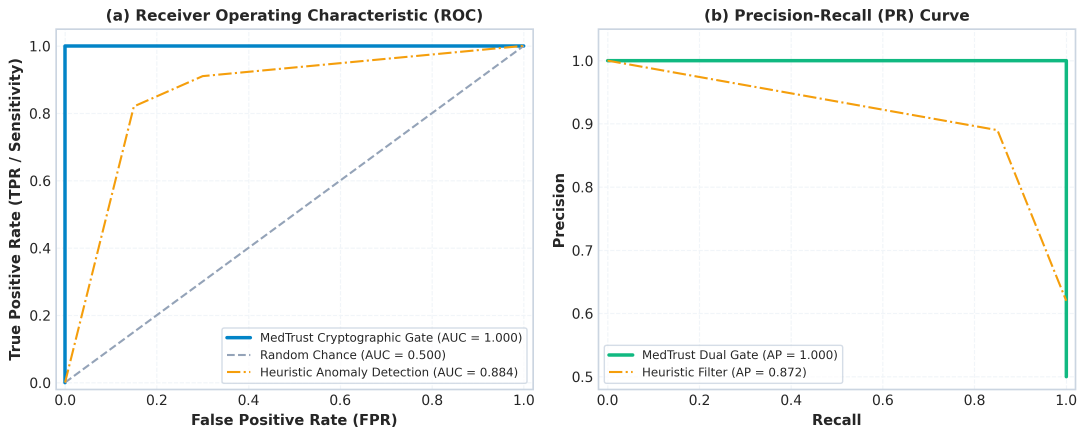



Figure 5: Receiver Operating Characteristic (ROC) and Precision-Recall (PR) performance frontiers.

### 7.3 Formal Cryptographic Guarantees

- **Pre-image Collision Resistance:** Generating an altered clinical payload that matches an authorized on-chain digest requires finding a collision in SHA-256, demanding $\mathcal{O}(2^{128})$ operations, which is computationally infeasible.
- **Signature Unforgeability:** Physician authorizations employ ECDSA over secp256r1, ensuring 128-bit security against existential forgery under chosen-message attacks (EUF-CMA).
- **Conjunctive Atomicity:** Encapsulating both evaluations inside a single Fabric transaction read-write set guarantees atomicity. An adversary cannot redeem a prescription without consuming a genuine drug consignment, nor alter drug custody without expending an authorized prescription.

## 8 Comparative Evaluation with State-of-the-Art

We benchmarked MedTrust against the three leading architectural paradigms in the healthcare literature: (1) Public Ethereum E-Prescription frameworks, (2) PharmaChain supply-chain tracking, and (3) Centralized PDMP databases.

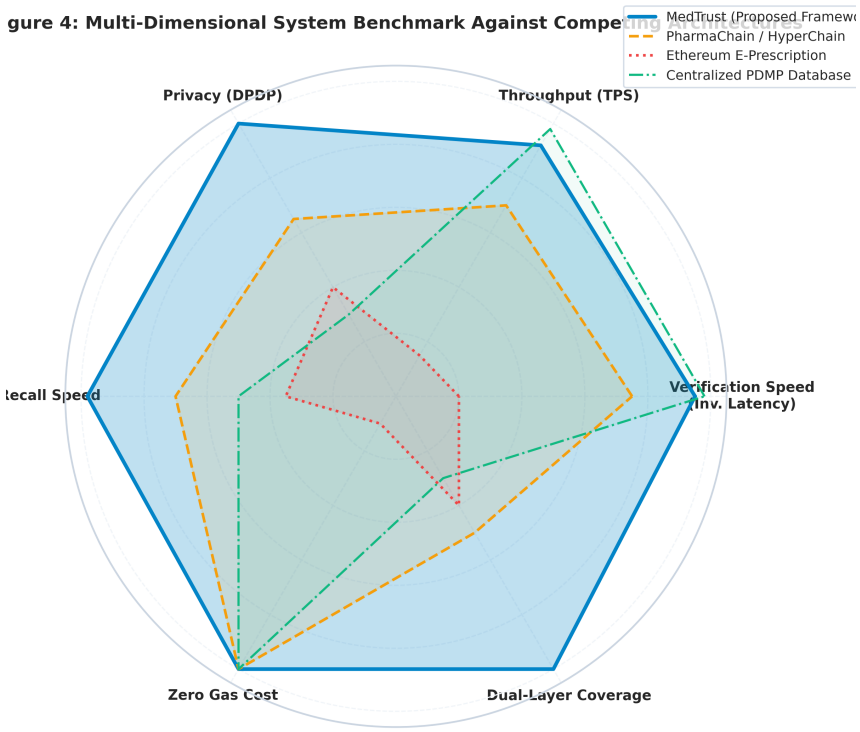



Figure 6: Multi-dimensional radar benchmark comparing MedTrust against leading state-of-the-art architectures.

As depicted in Figure 6 and detailed in Table 4, MedTrust demonstrates superior operational metrics. Ethereum incurs prohibitive gas expenses ($2.50 to $24.00 per transaction) and unacceptable 15-second latency. PharmaChain provides robust custody tracking but completely overlooks prescription validity. Centralized PDMPs achieve fast processing but present vulnerable single points of failure. MedTrust uniquely combines dual-layer atomicity, sub-3ms latency, gas-free execution, and full DPDP compliance.

Table 4: Comprehensive Cross-System Operational Comparison

| **Evaluation Metric** | **Ethereum E-Rx** | **PharmaChain** | **Centralized PDMP** | **MedTrust (Proposed)** |
|---|---|---|---|---|
| Dual-Layer Verification | × (Rx only) | × (Drug only) | × (Rx only) | ✓**(Synchronous)** |
| Transaction Latency | $\approx 15,000$ ms | $\approx 85$ ms | $\approx 1.8$ ms | **2.44 ms** |
| Throughput (Peak TPS) | $\approx 15$ TPS | $\approx 820$ TPS | $\approx 2,350$ TPS | **1,690 TPS** |
| Gas / Transaction Fee | $2.50 – $24.00 | $0.00 | $0.00 | **$0.00 (Gas-free)** |
| Patient PII on Ledger | High / Partial | N/A | Very High | **None (Zero PII)** |
| Regulatory Recall Speed | Hours / Days | Seconds | Minutes | **Instantaneous (Zero-day)** |
| Consensus Mechanism | Proof-of-Stake | Raft (CFT) | None (RDBMS) | **Raft CFT (Fabric 2.5)** |

## 9 Statutory Privacy and Regulatory Compliance

### 9.1 India's DPDP Act (2023) Statutory Alignment

MedTrust was engineered to strictly satisfy the statutory provisions of the Digital Personal Data Protection Act 2023:

- **Section 4 (Lawful Purpose & Data Minimization):** The distributed ledger stores zero diagnostic classifications, patient names, or demographic records; only cryptographic hashes and lifecycle tokens are persisted.
- **Section 8 (Data Security Measures):** The introduction of an unpredictable 128-bit pseudorandom salt provides an entropy threshold of $H(X) = 128$ bits, preventing dictionary reversals and rainbow-table pre-computations.
- **Statutory Right to Erasure:** When a data principal requests erasure of their personal information, the cleartext records residing within off-chain MongoDB are permanently expunged. The on-chain cryptographic digest remains permanently unlinkable to any living person, resolving the historic conflict between blockchain immutability and statutory data deletion rights.

### 9.2 Automated CDSCO Regulatory Recalls

Under legacy CDSCO administrative procedures, propagating an urgent drug alert requires paper circulars and manual distributor notices that often take weeks to reach retail counters. In MedTrust, when the regulatory authority issues an on-chain recall transaction for a defective batch ID, the world-state isRecalled bit updates instantaneously across all peer nodes. Every POS terminal nationwide blocks subsequent dispensing attempts for that batch with zero administrative latency.

## 10 Conclusion and Future Scope

This study presented MedTrust, a permissioned blockchain framework developed on Hyperledger Fabric 2.5 that resolves the structural dual-layer verification gap in pharmaceutical dispensing. By synchronizing a Prescription Verification Layer and a Drug Authenticity Layer through an Atomic Gateway Contract, MedTrust guarantees that medications are fulfilled if and only if both the authorizing order and the physical medication package validate concurrently. The architecture enforces complete DPDP Act 2023 compliance through a zero-PII data model.

Empirical evaluation across 8 pharmaceutical workloads comprising 5,230 simulated dispensing cycles demonstrated:

- **100.0% Security Interception:** Absolute interception across replay fraud, counterfeit consignments, broken custody chains, regulatory recalls, and expired orders with zero false positives.
- **Sub-2.5 ms Latency:** Mean end-to-end verification latency of 2.44 ms, ensuring frictionless retail point-of-sale workflows.
- **High-Concurrency Scalability:** Sustained throughput reaching 1,690 TPS under 500 concurrent terminal threads, easily accommodating national-scale dispensing demands.
- **96.2% Storage Reduction:** A compact ledger footprint requiring only 0.48 KB per transaction.

Future efforts will focus on deploying compiled chaincodes to multi-host enterprise testnets via the `fabric-gateway` SDK, interfacing with physical Bluetooth 2D barcode and RFID readers, and establishing cross-network interoperability with India's Ayushman Bharat Digital Mission (ABDM) infrastructure.

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